Two-Column Proofs — Practice — Answer Key

A worked example comes before each type. Fill in every missing statement and reason.

Name: _____ Date: _____

Type 1 Algebra Proofs

Watch Me — Worked Example

Given: $\frac{x}{2} + 3 = 7$ Prove: $x = 8$

1. $\frac{x}{2} + 3 = 7$ (Given) 2. $\frac{x}{2} = 4$ (Subtraction Property of Equality) 3. $x = 8$ (Multiplication Property of Equality)

1. Given: $4x + 9 = 25$ Prove: $x = 4$

Statements	Reasons
1. $4x + 9 = 25$	Given
2. $4x = 16$	Subtraction Property of Equality
3. $x = 4$	Division Property of Equality

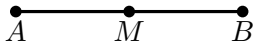
2. Given: $3(x - 1) = 2x + 5$ Prove: $x = 8$

Statements	Reasons
1. $3(x - 1) = 2x + 5$	Given
2. $3x - 3 = 2x + 5$	Distributive Property
3. $x - 3 = 5$	Subtraction Property of Equality
4. $x = 8$	Addition Property of Equality

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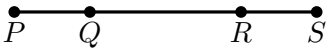
Type 2 Segment Proofs

Watch Me — Worked Example



Given: M is the midpoint of $\overline{AB}$. Prove: $AM = \frac{1}{2}AB$.
1. M is the midpoint (Given) 2. $AM = MB$ (Def. of midpoint) 3. $AM + MB = AB$ (Segment Addition) 4. $AM + AM = AB$ (Substitution) 5. $2AM = AB$ (Simplify) 6. $AM = \frac{1}{2}AB$ (Division Prop.)

3. Given: $PQ = RS$ Prove: $PR = QS$



Statements	Reasons
1. $PQ = RS$	Given
2. $QR = QR$	Reflexive Property
3. $PQ + QR = QR + RS$	Addition Property of Equality
4. $PQ + QR = PR,\ QR + RS = QS$	Segment Addition Postulate
5. $PR = QS$	Substitution Property

Type 3 Angle Proofs

Watch Me — Worked Example

Given: $\overrightarrow{BD}$ bisects $\angle ABC$. So $m\angle ABD = m\angle DBC$ — reason: **Definition of angle bisector**.
Then $m\angle ABD + m\angle DBC = m\angle ABC$ — reason: **Angle Addition Postulate**.

4. Given: $m\angle 1 = m\angle 3$ Prove: $m\angle ABD = m\angle CBE$ ($\angle ABD$ is made of $\angle 1$ and $\angle 2$; $\angle CBE$ is made of $\angle 2$ and $\angle 3$.)

Statements	Reasons
1. $m\angle 1 = m\angle 3$	Given
2. $m\angle 2 = m\angle 2$	Reflexive Property
3. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$	Addition Property of Equality
4. $m\angle 1 + m\angle 2 = m\angle ABD$	Angle Addition Postulate
5. $m\angle 2 + m\angle 3 = m\angle CBE$	Angle Addition Postulate
6. $m\angle ABD = m\angle CBE$	Substitution Property

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Type 4 Find the Mistake

Watch Me — Worked Example

A student wrote "$\angle 1 \cong \angle 2$ — Reason: they look the same." That is not a reason. Use a given, definition, property, postulate or theorem.

5. Step: "$2x = 10 \Rightarrow x = 5$. Reason: Subtraction Property." Fix the reason.

Division Property of Equality

6. HARD Step: "$AB = AB$. Reason: Symmetric Property." Fix the reason.

Reflexive Property

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